MATH FOUNDATIONS

Graph Theory Visual Guide

Paths, flows, and colorings — the math behind networks, schedules, and systems.

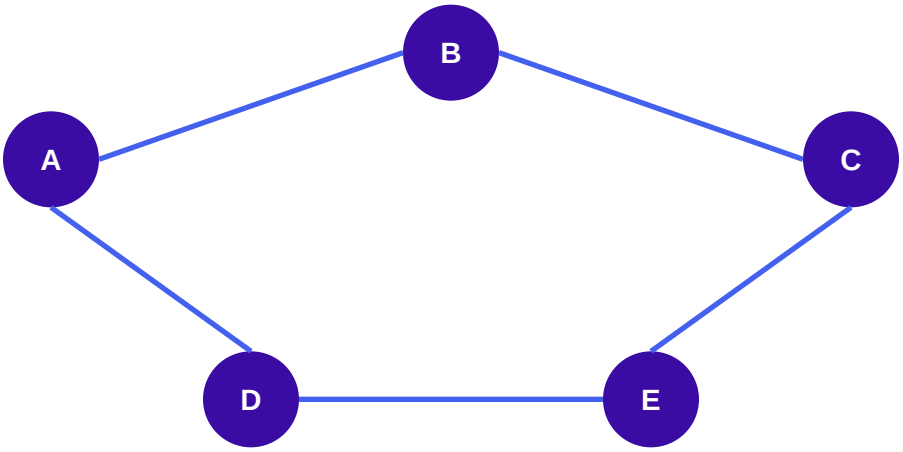
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Graphs 101: Vertices + Edges

Everything is dots and lines: social nets, road maps, dependency trees.



UNDIRECTED

edges have no direction — friendships

DIRECTED

edges point — follows, dependencies

Weighted edges add costs/distances; then shortest-path and flow problems come alive.

STEP 2/6

Paths, Cycles, Connectivity

The vocabulary every graph algorithm speaks.

Path	sequence of vertices with no repeats — A-B-D
Cycle	a path that returns to its start — A-B-D-A
Connected	a path exists between every pair of vertices
DAG	directed + acyclic — task dependencies, build order

If your dependency graph has a cycle, nothing can build. Topo-sort a DAG to find the order.

STEP 3/6

Shortest Path: BFS and Dijkstra

Unweighted? BFS. Weighted non-negative? Dijkstra.

BFS

layer by layer from the source — first visit = shortest (unweighted)

DIJKSTRA

always expand the closest unsettled vertex (priority queue)

Greedy + priority queue = O((V+E) log V). Negative edges? Bellman-Ford instead.

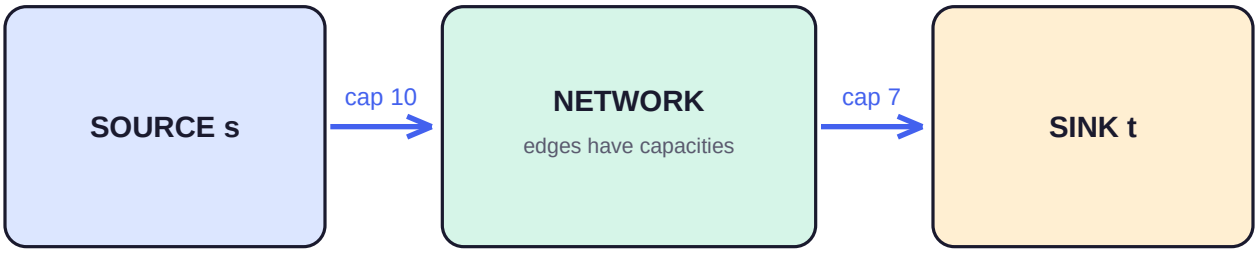
USE IN SYSTEMS

routing tables, dependency resolution, game AI pathfinding

STEP 4/6

Max Flow / Min Cut

Push as much as possible from source to sink through capacitated edges.



MIN-CUT = MAX-FLOW

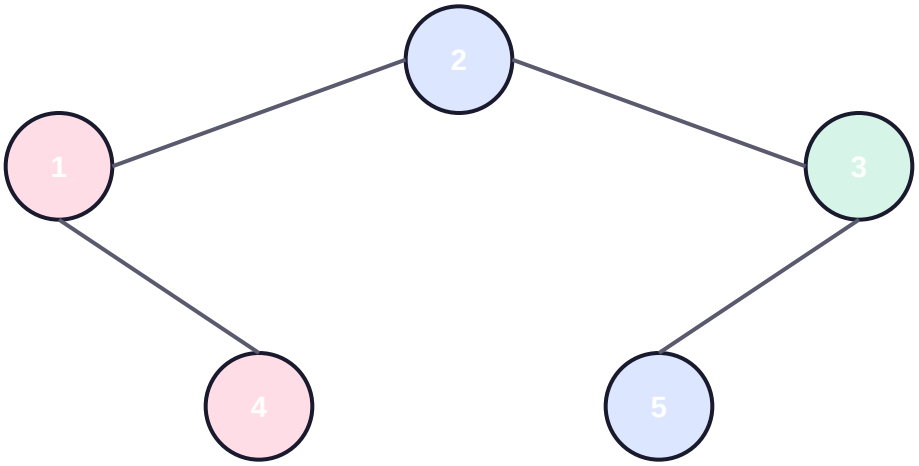
the cheapest set of edges whose removal disconnects s from t

Bipartite matching, image segmentation, project selection — all max-flow in disguise.

STEP 5/6

Graph Coloring

Adjacent vertices get different colors — scheduling's secret weapon.



CHROMATIC NUMBER

minimum colors needed — exam slots, register allocation, map coloring

Optimal coloring is NP-hard, but greedy orderings work well in practice.

STEP 6/6

Graphs in Real Systems

If it connects, it is a graph problem.

Dependency DAGs

build systems, package managers — topo-sort for order

Computer networks

routers as vertices; OSPF runs Dijkstra on the topology

PageRank

eigenvector centrality: important pages are linked by important pages

Social graphs

communities, influence, recommendations via graph algorithms

STEP ✓I✓

Key Takeaways

Graph Theory — Remember

- Vertices plus edges model anything connected: networks, dependencies, maps.
- Paths and cycles are the vocabulary; DAGs give you build order via topological sort.
- BFS for unweighted shortest path; Dijkstra with a priority queue for weighted.
- Max flow equals min cut — matching, segmentation, and selection reduce to flow.
- Coloring assigns resources so neighbors differ; greedy works well in practice.
- Routing, PageRank, builds, and social graphs are all graph algorithms in production.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev